IDEATHON

Automated PAR Cue

Radar data to pilot guidance, without a controller

Team Beetiful

EUROAVIA Ideathon 2026 – Advionics challenge

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Oostende

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1 Introduction

A Precision Approach Radar (PAR) shows a controller the distance, lateral offset and height offset of an aircraft on final approach. The controller reads these values and talks the pilot down in low visibility [7, 3, 4]. FAA Order JO 7110.65BB asks for one transmission about every 5 s [2]. Advionics asks us to automate this chain with no controller. Radar data goes in, and heading and glidepath corrections go out as a clear cue to the pilot [1].

We chose the HYBRID direction (Figure 1). A display updates with every radar plot, and voice speaks only for important changes. A 5 s voice rhythm is a limit of the human talk-down, and a display does not block the radio. The words are FAA words, so a pilot who knows a PAR talk-down can read them at once.

We assume a working radar, an aircraft established on final and a decision height (DH) of 200 ft. This is an educational concept on simulated and recorded data. It is not approved or ready for operational flight use.

2 Synthetic data

2.1 Purpose and limits

We found no measured PAR plots. A recorded track also cannot test the cue, because a recorded aircraft does not react to it. So we built a closed loop: radar plots feed the cue, a pilot model flies the cue, and the aircraft moves. Faults start at a known time and repeat with the same seed. The model tests the logic. It does not predict how a real pilot performs.

2.2 Basic equations

Radar error uses the ICAO Annex 10 limits [8]. R is the distance from the antenna; y and z are the lateral and vertical deviation. We assume that 1σ is half of the limit:

$$\sigma_{\text{az}} = \tfrac{1}{2}\max(0.006R + 0.1|y|,\ 9\,\text{m}), \qquad \sigma_{\text{el}} = \tfrac{1}{2}\max(0.004R + 0.1|z|,\ 6\,\text{m}). \tag{1}$$

The aircraft is a point mass with airspeed V, heading ψ, sink rate s, crosswind W_c and vertical gust w:

$$\dot{y} = V\sin\psi + W_c, \qquad \dot{z} = w - s. \tag{2}$$

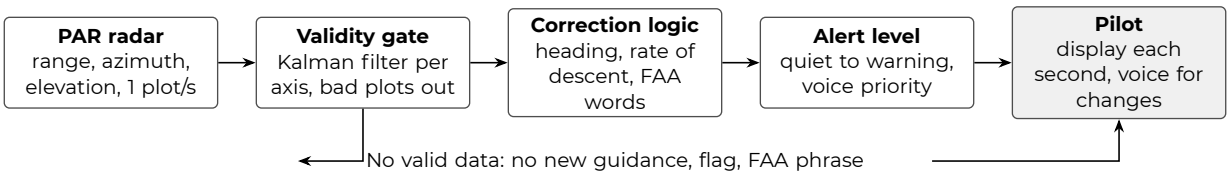


Figure 1. Radar to pilot flow. The link from the ground to the cockpit is assumed.

Table 1. The model covers the full chain at low order. Simulation results hold only inside this scope.

Element	In the model	Left out
Radar geometry	Antenna 915 m back and 120 m aside, 1 plot/s, 9 NM, 20° sector [8]	Beam scan, clutter, rain
Radar error	Annex 10 limits, half of the variance correlated over 5 s; 0.05° multipath term [both assumed]	Measured PAR error, real terrain
Plot loss	95 % detection, 90 % as a test case	Long burst losses
Wind	Log-law profile, crosswind up to 10 m/s [12]	Microburst, wind-shear events
Turbulence	Dryden, light (15 kt) and moderate (30 kt) [11]	Severe turbulence, wake
Aircraft	Point mass, 130 kt C-130H or 137 kt A320 [13], 4 s lag	6-DOF dynamics, speed change, flare
Pilot	2 s reaction [no PAR source; analogy from EGPWS response], 1.5°/s turns, height loop 0.25 rad/s [14], 1° pitch limit	Workload, fatigue, startle, pilot error
Faults, link	Delay, jump, invalid flag, radar stop, link on or off	Spoofing, link latency, bit errors
Procedure	Starts on final at 5 NM	Vectors, begin-descent call, mile calls by voice

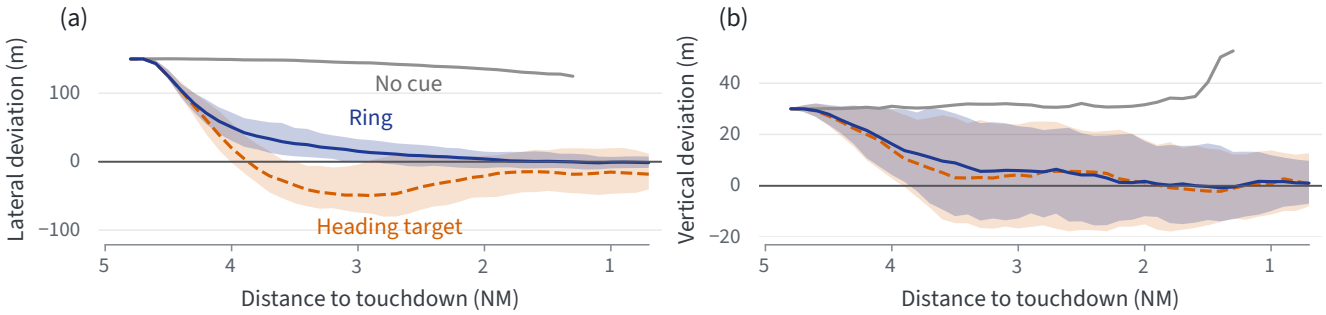


Figure 2. Both cue laws remove the start offset. The ring also removes the overshoot. Lines are medians and bands are 5 to 95 % of 100 runs. The no-cue line ends where most runs leave the limits.

Wind grows with height h from its 20 ft value W_{20} (h and $z_0 = 0.15$ in ft). Gusts follow the low-altitude Dryden form with turbulence level W_t, 15 kt when light [11, 12]:

$$W(h) = W_{20}\,\frac{\ln(h/z_0)}{\ln(20/z_0)}, \qquad \sigma_w = 0.1\,W_t. \tag{3}$$

2.3 Closed-loop result

The test follows the brief. The aircraft starts at 5 NM, 150 m right and 30 m high, with 3 m/s crosswind and light turbulence. We ran 100 seeds. The first pilot model flies a rounded heading target that changes at most every 5 s, as a controller's heading does. The second flies the marker into the display ring (Section 4).

With the heading target, 99 of 100 runs reach DH, with a 95 % error of 37 m lateral and 11 m vertical. With the ring, 99 of 100 reach DH with 11 m and 9 m. That ring pilot reads drift from the marker with no error, so 11 m is a best case. With no cue, 1 of 100 reaches DH. The heading target gives 95 of 100 with a 4 s reaction and 96 at 90 % detection. It gives 100 with a radar twice as good and 77 in moderate turbulence (ring: 76). A 300 m start at 5 m/s crosswind gives 65.

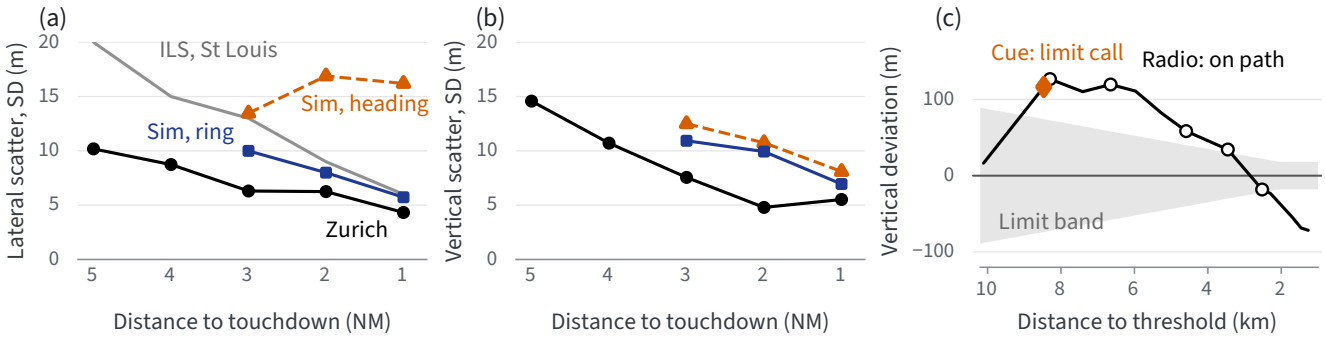


Figure 3. Simulated scatter is wider than real scatter. (a, b) Scatter between flights: 1000 recorded tracks, 100 simulated runs and published ILS data [17]. (c) Smolensk 2010: track from 13 report points, ±0.5° limit, the five radio position calls (circles) and the first limit call of the cue in replay (diamond).

3 Test on recorded data

3.1 Zurich: agreement and replay

We used 1000 recorded tracks on Zurich runway 14 (ADS-B, October and November 2019) [15]. They are the first 1000 tracks that passed our filter, not a random sample. For this comparison the simulation starts on the path, with no steady wind, at A320 speed. Speed is an input (137 kt; recorded median 141 kt). Sink rate is a result: 726 ft/min against 747 ft/min recorded, with 5 to 95 % ranges of 640–849 and 667–842 ft/min.

With the ring, simulated lateral scatter is 1.3 to 1.6 times the recorded value, and vertical scatter 1.3 to 2.1 times. With the heading target, lateral scatter is 2.1 to 3.7 times the recorded value. A perfect radar does not reduce it (13 to 18 m), so the cause is the heading steps. A perfect radar does reduce vertical scatter to 4 to 5 m, so PAR accuracy sets the vertical result. These aircraft flew ILS, and the simulated aircraft flew the cue, so the two scatters are not a like-for-like test. We also removed one height constant per flight, because ADS-B height differs by aircraft. Recorded vertical scatter is therefore a lower bound.

Each track then went through the Annex 10 radar model and the cue. The aircraft did not react to the cue. No pilot saw it. Of 1000 tracks, 985 stayed quiet, 5 reached advise, 4 caution and 6 warning. On 98.5 % of tracks the cue gave no guidance voice, and 275 tracks ended before DH. We did not check whether the 6 warnings were real excursions. At 90 % detection, 8 tracks ended in a false RADAR LOST.

Recorded data changed the design. With no noise margin and no frame count, the closure flag fired on 397 of 1000 tracks. Simulation had not shown this. With both it fires on 6. This is not an independent result, because we tuned the flag on the first 300 of these tracks and on the Smolensk track. Of 66 go-arounds labelled on this runway [16], we could cut 22 out of the tracks. The cue spoke before the climb in 11, with a median lead of 15 s.

3.2 Smolensk 2010: a radar talk-down

This is the one event we found with a radar talk-down and stated deviations [18]. We use deviations and calls only, with no finding on blame. The radio said “on course, on glide path” five times outside the on-path band. The first correction came 5.5 s before the first tree strike. In replay, our limit call would have come 88.5 s before it (median, 200 of 200 noisy runs). With the limit switched off, “Coming down rapidly” would have come 16.5 s before it, in 167 of 200 runs. If the radar itself was wrong, as the report suspects, the cue would inherit the error.

4 The cue on the display

4.1 Marker logic

ICAO treats the axes differently [9]. In azimuth the pilot gets a heading. In elevation the pilot corrects from position information. The law follows this and uses radar position only. Three designs with a radar-derived rate failed (0 of 100 runs reached DH), because a rate from 1 Hz plots is too noisy. With ground speed V_g, range r and path angle γ, the look-ahead time T, the aim track χ_r and the aim sink rate s_r are:

$$T = \min\!\Big(20\,\mathrm{s},\ \max\big(\tfrac{r}{2V_g},\ 8\,\mathrm{s}\big)\Big), \quad \chi_r = -\arctan\frac{y}{V_g T}, \quad s_r = V_g \tan\gamma + \frac{z}{T}. \tag{4}$$

Corrections are capped at 10° (5° inside 3 NM) and 1.2 m/s. We use symbols that pilots already know (Figure 4):

- **Diamonds: where the path is.** Scales as on an ILS, 1 dot = 1.25° lateral and 0.35° vertical.
- **Marker: where you go.** Real track and flight path angle from aircraft sensors, not from the radar. Wind needs no term.
- **Ring: where to go.** The point on the path T seconds ahead, at χ_r and s_r. It turns green with the marker inside.

With the marker in the ring, $\dot{y} = -y/T$, so the offset decays with time constant T. An aircraft with no flight path marker cannot use the ring. It can use the diamonds, the heading and the FAA words.

4.2 Alert levels and voice

Below 1 dot the level is quiet. At 1 dot it is advise (stable approach limit [19]). At 1.3 dots it is caution (EGPWS Mode 5 soft alert [20]). In a missed approach state it is warning. The limits ±0.5° and ±2° come from one report [18]. The FAA prints no number, and a real system takes them per runway. A call repeats only for 20 % more deviation. The cue ranks below every aircraft alert [6]. With an aircraft alert sounding, the cue is silent and a warning waits. With

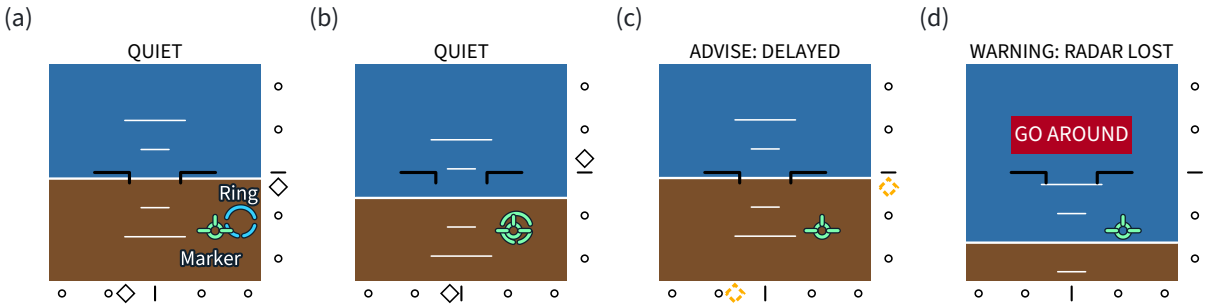


Figure 4. The symbols show the state before the pilot reads any text. (a) Off path: marker outside the ring. (b) On path: marker in the green ring. (c) Data 4 s old: hollow diamonds, no ring. (d) Radar lost: no diamonds, no ring. Replay frames; horizon position is schematic.

Table 2. Each fault has one display state and one call. The pilot never sees old data without a mark. 3 s and 4 NM are UK values [10]; 5 s is the longest ICAO silence [9] and the FAA lost communications interval.

Event	Told after	Display	Voice	Basis
Plots late	3 s	Diamonds hollow, ring off, no new heading	“Radar data delayed”	RA 3291
Bad plots (jump, invalid)	2 plots	Diamond of that axis off, back after 3 good plots	“Radar data invalid”	Design value
Radar contact lost	3 s (5 s beyond 4 NM)	All guidance off, latched	“Radar contact lost. If runway or approach lights not in sight, execute missed approach.”	JO 5-10-14c
Elevation failed	2 plots	All guidance off, latched	“No glidepath information available. If runway…”	JO 5-12-10
Outside limits	3 s	All guidance off, latched	“Too low for safe approach. If runway…”	JO 5-10-14a
Link silent	5 s	All guidance off	“Guidance link lost. Apply lost communications procedure.” Not a missed approach	JO 5-10-4
Any fault below DH	–	Diamond off, advisory text	None	JO 5-10-14, 5-12-7

a transmission on, a routine call is dropped. The display never waits. Voice repeats the display words [5].

5 When data goes wrong

With no valid data, the cue gives no new guidance and no extrapolation (Table 2). A missed approach call is the reason, then the condition [2]. The pilot decides. The short GO AROUND flag on the mock-up points to that call. None of these calls applies after DH.

In all 100 seeded runs, the cue told a 4 s delay and a radar stop after 3 s. It told a 400 m jump after 1 s in 90 runs. No frame showed guidance without valid data. The weak point is a wrong but plausible plot. Far out, a jump below 200 m was not caught. Near the runway, a jump below 150 m was mostly not caught. Such a jump moved the aircraft by up to 66 m (95 %).

6 Limits and next step

No real pilot has flown this cue, so we claim no drop in workload. Voice less often than every 5 s departs from the FAA rhythm and needs a pilot trial. Radar to cockpit needles is not new:

Navy ACLS Mode II does this [21, 2]. Our work is the logic behind the display: FAA fault rules, a silent path and a voice that yields to aircraft alerts. We tested it in a replay of 1000 recorded tracks.

For Advionics the airborne change is one display page and one data link, and the radar stays. The work fits its RF, embedded software and test skills [22]. The next step is a pilot-in-the-loop simulator trial. Recorded PAR plots, link latency tests and an independent height check follow.

7 References

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Table 3. FAA rules behind the design. The right column is our use of them.

FAA source	What it says	What our system does
JO 7110.65BB 5-12-4a	Inform on glidepath and on course, and of any deviation; transmit about every 5 s. Words: ON GLIDEPATH, ON COURSE, SLIGHTLY/WELL ABOVE/BELOW GLIDEPATH, SLIGHTLY/WELL LEFT/RIGHT OF COURSE	Same words on the display each second. Voice is rarer on purpose: this departs from the rule and needs a trial
JO 7110.65BB 5-12-4b	Trend information as required, with RAPIDLY or SLOWLY	Trend words from a 10 s fit, for words only, never for control
JO 7110.65BB 5-10-14	If the runway environment is not in sight, instruct a missed approach when safety limits are exceeded, position is in doubt, or radar contact is lost or a malfunction suspected. Reason first. Not after DH on a PAR approach	Reason, then “if runway or approach lights not in sight, execute missed approach”. No such call after DH
JO 7110.65BB 5-10-4	Tell the pilot what to do if radio is lost for 5 s on a PAR final. ACLS Modes I, IA and II: 5 s after loss of data link	Link silent 5 s: the pilot applies the lost communications instruction. Not a missed approach
JO 7110.65BB 5-12-7, 5-12-8	Position information until over the threshold. Below DH it is advisory only (14 CFR 91.175 is the pilot’s). Guidance ends over the threshold	Position stays on below DH, marked ADVISORY ONLY, and ends over the threshold
JO 7110.65BB 5-12-10	Elevation failure: stop PAR instructions; take over visually or execute a missed approach. NO GLIDEPATH INFORMATION AVAILABLE	Ends guidance after 2 flagged plots, with this phrase
AIM 5-4-11	A PAR pilot gets headings. For elevation the pilot gets “slightly” or “well” and adjusts rate of descent. Outside safety limits: missed approach unless the runway environment is in sight	Heading in azimuth, information in elevation. A limit must be exceeded for 3 s before the call
AC 25.1322-1 para 12, App. 2	Design to avoid false and nuisance alerts. Voice should match the displayed text	Silent on the path. Voice uses the display words
FAA HF guidance 4.3 (TSO-C151c Table 4-2)	Published priority order of TAWS alerts	The cue ranks below every line of that list